Graphic Era HILL UNIVERSITY
Established by an Act of the State Legislature of Uttarakhand (Adhiniyam Sankhya 12 of 2011)
University under section 2(f) of UGC Act, 1956

Mid Term (Odd) Semester Examination October 2024

Roll no.....

Name of the Course and semester: B.Tech, ME, III semester
Name of the Paper: Thermal Engineering
Paper Code: TME 308

Time: 1.5 hour

Maximum Marks: 50

Note:

(i) Answer all the questions by choosing any one of the sub questions
(ii) Each question carries 10 marks.

Q1. (10 Marks)

a. What do you understand by thermodynamic equilibrium? **CO 1**

OR

b. What do you understand by macroscopic and microscopic viewpoints? **CO 1**

Q2. (10 Marks)

a. Assume that the pressure p and the specific volume v of the atmosphere are related according to the equation $pv^{1.4} = 2.3\times105$, where p is in N/m^2 abs and v is in m^3/kg. The acceleration due to gravity is constant at 9.81 m/s^2. What is the depth of atmosphere necessary to produce a pressure of 1.0132 bar at the earth's surface? Consider the atmosphere as a fluid column. **CO 1**

OR

b. A pump discharges a liquid into a drum at the rate of 0.032 m^3/s. The drum, 1.50 m in diameter and 4.20 m in length, can hold 3000 kg of the liquid. Find the density of the liquid and the mass flow rate of the liquid handled by the pump.
CO 1

Q3. (10 Marks)

a. Explain concept of continuum. **CO 1**

OR

b. What do you understand by point function and path function? **CO 2**

Q4. (10 Marks)

a. A turbine operates under steady flow conditions, receiving steam at the following state: Pressure 1.2 MPa, temperature 188°C, enthalpy 2785 kJ/kg, velocity 33.3 m/s and elevation 3 m. The steam leaves the turbine at the following state: Pressure 20 kPa, enthalpy 2512 kJ/kg, velocity 100 m/s, and elevation 0 m. Heat is lost to the surroundings at the rate of 0.29 kJ/s. If the rate of steam flow through the turbine is 0.42 kg/s, what is the power output of the turbine in kW? **CO 2**

OR

b. It is required to melt 5 tonnes/h of iron form a charge at 15°C to molten metal at 1650°C. The melting point is 1535°C, and the latent heat is 270kJ/kg. the specific heat in solid state is 0.502kJ/kgK and in liquid state 29.93/atomic weight kJ/kgK. If an electric furnace has 70% efficiency, find the kW rating needed. If the density in molten state is 6900 kg/m3 and the bath volume is three times the hourly melting rate, find the dimensions of the cylindrical furnace if the length to diameter ratio is 2. The atomic weight of iron is 56. **CO 2**

Q5. (10 Marks)

a. Explain PMM1 and PMM2. **CO 2**

OR

b. Derive the steady flow energy equation for a single stream entering and a single stream leaving a control volume and explain the various terms in it. **CO 2**

Scanned with ACE Scanner